HPDP Project Brief

Hyperspectral Payload Data Processor: an FPGA model of a satellite's onboard data chain, from raw sensor data to a bit-exact image at the ground

Document	HPDP-BRF-001, revision C
Date	2026-10-23
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Hardware	AMD ZCU106 evaluation board, Zynq UltraScale+ XCZU7EV-2
Standards	CCSDS 123.0-B-2 (compression), 133.0-B (space packets), 732.0-B (AOS frames), 131.0-B (channel coding)
Duration	10 to 15 October 2026, eight milestones, 51 commits; Ethernet upload tested on the board 23 October
Status	All milestones (M0 to M7) complete; all 102 requirements met

At a glance	
What	The complete onboard data chain of a hyperspectral imaging satellite, running in FPGA logic, with a fiber link to a ground receiver
Throughput	76.8 Msamples/s (the mission rate) for 10 minutes, no line lost
Integrity	Every scene rebuilt bit for bit at the ground, at bit error rates up to 10^{-3}
Link	3.4×10^{12} bits over the fiber with zero errors
Radiation	Configuration upsets corrected in 153 ms or less; no silent error in protected logic

Further documents	
HPDP-PRC-001	Process Description: how the project was done, step by step, with the procedures to rebuild and run it (Simplified Technical English)
HPDP-RPT-001	Results Report: all test results, charts and the requirement compliance
HPDP-GDE-001	Learning Guide: the HPDP taught from the basics to the full design, with check questions (Simplified Technical English)
SPEC.md	The specification (v1.0)
docs/	project_report.md and the engineering notes of each milestone (m0 to m7)

1 What the HPDP is

A hyperspectral satellite photographs the ground in hundreds of narrow colour bands. Its camera produces far more data than its radio downlink can carry, the downlink corrupts bits, and radiation in orbit flips bits inside the electronics. The **Hyperspectral Payload Data Processor (HPDP)** is a working model of the hardware that solves these problems on board, built on a single AMD ZCU106 FPGA board.

Raw sensor data enters the FPGA, is cleaned up (radiometric correction), compressed without loss (CCSDS 123.0-B-2), packed into space packets and transfer frames, protected with Reed-Solomon coding, and sent at 1.25 Gbaud over an optical fiber. A ground receiver on the same board recovers the frames, and a PC ground station rebuilds the original image and checks it bit for bit.

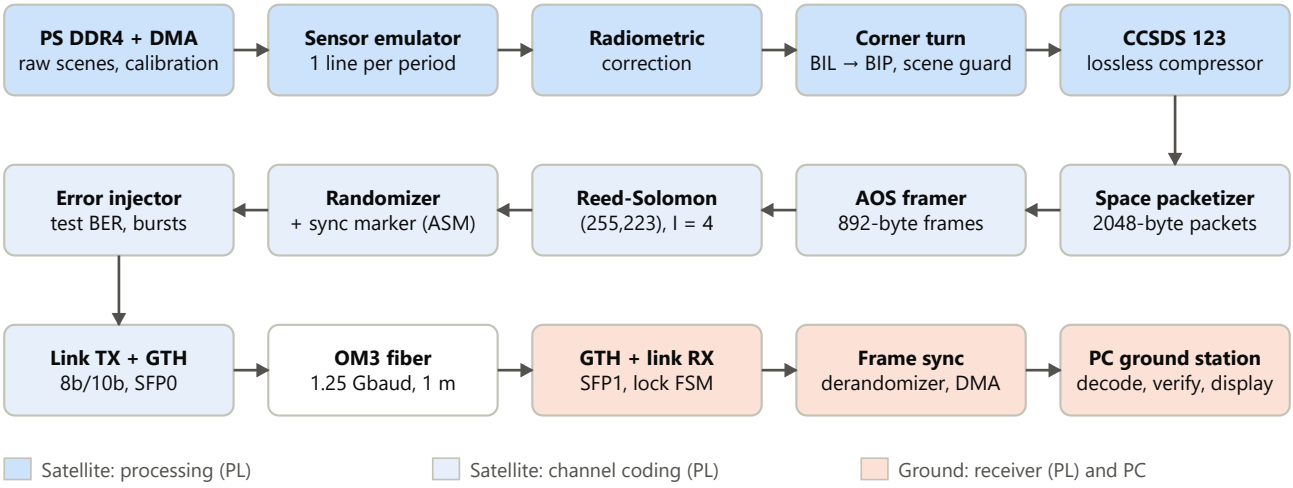


Figure 1: The HPDP data chain. Everything up to the frame synchronizer runs in the FPGA logic; the Arm processor on the same chip runs the control software.

Why compression is essential

At the reference mission rate of 76.8 million 12-bit samples per second, the raw data (922 Mbit/s) does not fit the link's packet capacity (857 Mbit/s). Lossless compression brings it down to roughly half, with room to spare even for the hardest real scene tested.

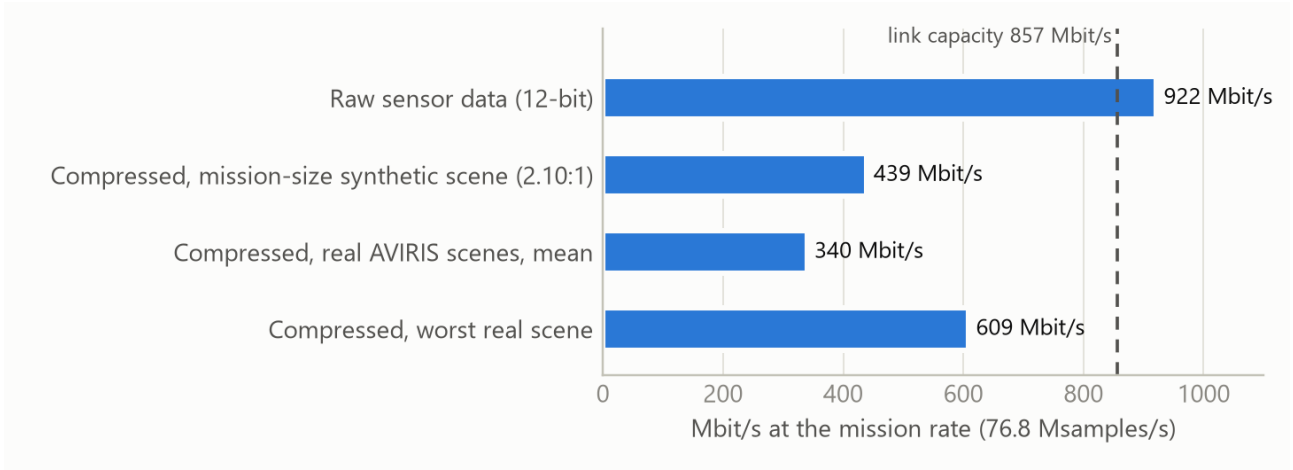


Figure 2: Data rate at the mission rate against the link capacity. The raw data does not fit; compressed data does.

2 Key results

Area	Result	Requirement
Throughput	76.8 Msamples/s (922 Mbit/s raw) for 10 minutes with no dropped line; the compressor alone runs at 250 Msamples/s	VER-08
End-to-end integrity	Every scene rebuilt bit-exactly at the ground with injected bit error rates of 0, 10^{-4} and 10^{-3}	VER-07
Optical link	PRBS-31 over the fiber: 3.4×10^{12} bits, zero errors (BER below 8.9×10^{-13}); open eye scans in both directions	VER-06, LNK-10
Compression, real data	63 public AVIRIS and Hyperion scenes, all lossless, 2.3:1 to 3.1:1 on average per dataset	SPEC §11
Radiation mitigation	Configuration scrubbing, ECC on all memories, triple redundancy, watchdogs; 251 upsets injected on the board, all handled	RAD-01 to 07
Timing and resources	Met on all 11 clocks (250 MHz data path); 17 % of the LUTs and 58 % of the UltraRAM	VER-09
Verification	150 model tests, 87 RTL tests, all 864 compressor configurations bit-exact, 1300-trial fault campaign	VER-01 to 10

How it was built

- **Specification first.** A 102-requirement specification with identifiers, milestones and acceptance criteria came before any hardware.
- **Model first.** Bit-exact Python models of every block generate the test vectors and the expected results.
- **Bit-exact verification.** Every RTL test compares every output bit with the model, with random stalls on every interface.
- **Safe board tests.** All hardware runs load over JTAG only; the board's flash was never written.
- **Tools.** AMD Vivado and Vitis 2025.2, Python, SystemRDL with PeakRDL and Git. The specification estimated about 31 part-time weeks; the work took six days.

Milestones

Milestone	Content	Result
M0	Tool flow, board bring-up	DMA loopback at 3865 MB/s
M1	Python reference models	150 tests
M2	CCSDS 123 compressor RTL	Bit-exact, 250 MHz
M3	Sensor emulator, correction, corner turn	Bit-exact, 250 MHz
M4	First compression on the board	15 scenes bit-exact
M5	Downlink chain and optical link	Zero-error fiber link; scenes bit-exact
M6	PC ground station	Decode, verify, display; Ethernet upload
M7	Radiation mitigation	Scrubbing, ECC, TMR, watchdogs

What remains

Nothing that the specification requires. The compressed output is byte-identical with an independent CCSDS 123.0-B-2 implementation (the NTNU SmallSat Lab's model) on 944 images, the board's scenes among them. Optional: a decoder-side check with the CNES tools or hcomp.